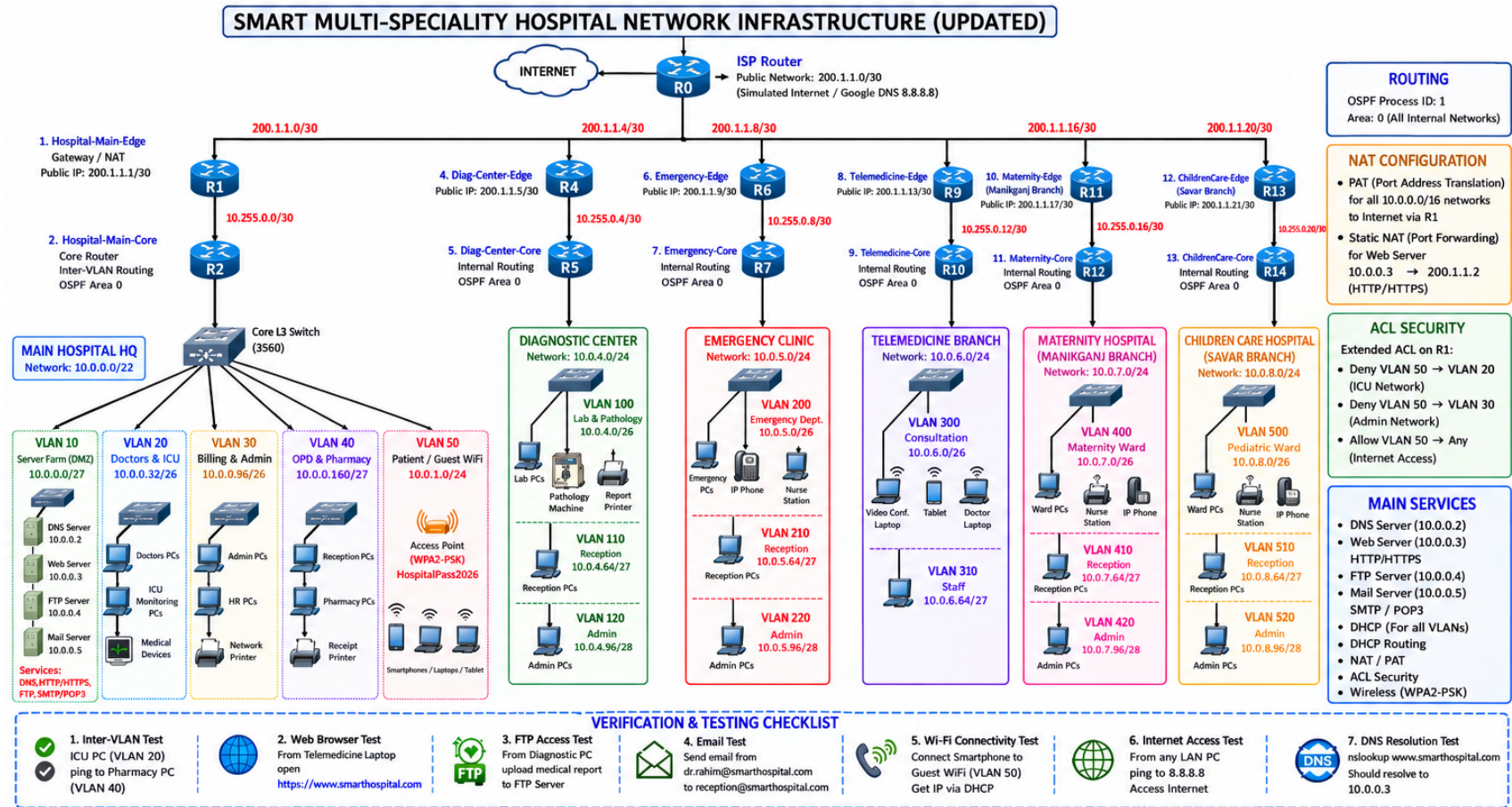


SMART MULTI-SPECIALITY HOSPITAL NETWORK INFRASTRUCTURE

Network Topology Diagram (Updated & Verified)



IP ADDRESSING SUMMARY		
Site / Branch	Network Block	VLANs (Departments)
Main Hospital HQ	10.0.0.0/22	10, 20, 30, 40, 50
Diagnostic Center	10.0.4.0/24	100, 110, 120
Emergency Clinic	10.0.5.0/24	200, 210, 220

Site / Branch	Network Block	VLANs (Departments)
Telemedicine Branch	10.0.6.0/24	300, 310
Maternity Hospital (Manikganj)	10.0.7.0/24	400, 410, 420
Children Care Hospital (Savar)	10.0.8.0/24	500, 510, 520

WAN / TRANSIT NETWORK		
WAN Links Network: 10.255.0.0/24		
Point-to-Point Links (/30):		
R1-R2: 10.255.0.0/30	R4-R5: 10.255.0.4/30	R6-R7: 10.255.0.8/30
R9-R10: 10.255.0.12/30	R11-R12: 10.255.0.16/30	R13-R14: 10.255.0.20/30

HIERARCHICAL VLSM ADDRESSING PLAN

Subnets allocated in descending order of host requirements within each branch block

1. Main Hospital HQ (Allocated Block: 10.0.0.0/22)

Order	VLAN	Department	Req. Hosts	Prefix	Network Address	Usable IP Range	Broadcast	Max Hosts
1	50	Guest Wi-Fi	200	/24	10.0.0.0	10.0.0.1 – 10.0.0.254	10.0.0.255	254
2	20	Doctors & ICU	50	/26	10.0.1.0	10.0.1.1 – 10.0.1.62	10.0.1.63	62
3	30	Billing & Admin	40	/26	10.0.1.64	10.0.1.65 – 10.0.1.126	10.0.1.127	62
4	40	OPD & Pharmacy	25	/27	10.0.1.128	10.0.1.129 – 10.0.1.158	10.0.1.159	30
5	10	DMZ	10	/28	10.0.1.160	10.0.1.161 – 10.0.1.174	10.0.1.175	14

Unused Space: 10.0.1.176 – 10.0.3.255

2. Diagnostic Center (Allocated Block: 10.0.4.0/24)

Order	VLAN	Department	Req. Hosts	Prefix	Network Address	Usable IP Range	Broadcast	Max Hosts
1	100	Laboratory	50	/26	10.0.4.0	10.0.4.1 – 10.0.4.62	10.0.4.63	62
2	110	Reception	20	/27	10.0.4.64	10.0.4.65 – 10.0.4.94	10.0.4.95	30
3	120	Administration	10	/28	10.0.4.96	10.0.4.97 – 10.0.4.110	10.0.4.111	14

Unused Space: 10.0.4.112 – 10.0.4.255

3. Emergency Clinic (Allocated Block: 10.0.5.0/24)

Order	VLAN	Department	Req. Hosts	Prefix	Network Address	Usable IP Range	Broadcast	Max Hosts
1	200	Emergency Ward	55	/26	10.0.5.0	10.0.5.1 – 10.0.5.62	10.0.5.63	62
2	210	Reception	20	/27	10.0.5.64	10.0.5.65 – 10.0.5.94	10.0.5.95	30
3	220	Administration	10	/28	10.0.5.96	10.0.5.97 – 10.0.5.110	10.0.5.111	14

Unused Space: 10.0.5.112 – 10.0.5.255

HIERARCHICAL VLSM ADDRESSING PLAN (CONTINUED)

4. Telemedicine Branch (Allocated Block: 10.0.6.0/24)

Order	VLAN	Department	Req. Hosts	Prefix	Network Address	Usable IP Range	Broadcast	Max Hosts
1	300	Consultation	40	/26	10.0.6.0	10.0.6.1 – 10.0.6.62	10.0.6.63	62
2	310	Staff	20	/27	10.0.6.64	10.0.6.65 – 10.0.6.94	10.0.6.95	30

Unused Space: 10.0.6.96 – 10.0.6.255

5. Manikganj Maternity Hospital (Allocated Block: 10.0.7.0/24)

Order	VLAN	Department	Req. Hosts	Prefix	Network Address	Usable IP Range	Broadcast	Max Hosts
1	400	Maternity Ward	50	/26	10.0.7.0	10.0.7.1 – 10.0.7.62	10.0.7.63	62
2	410	Reception	20	/27	10.0.7.64	10.0.7.65 – 10.0.7.94	10.0.7.95	30
3	420	Administration	10	/28	10.0.7.96	10.0.7.97 – 10.0.7.110	10.0.7.111	14

Unused Space: 10.0.7.112 – 10.0.7.255

6. Savar Children Care Hospital (Allocated Block: 10.0.8.0/24)

Order	VLAN	Department	Req. Hosts	Prefix	Network Address	Usable IP Range	Broadcast	Max Hosts
1	500	Pediatric Ward	50	/26	10.0.8.0	10.0.8.1 – 10.0.8.62	10.0.8.63	62
2	510	Reception	20	/27	10.0.8.64	10.0.8.65 – 10.0.8.94	10.0.8.95	30
3	520	Administration	10	/28	10.0.8.96	10.0.8.97 – 10.0.8.110	10.0.8.111	14

Unused Space: 10.0.8.112 – 10.0.8.255

Why This Is Good VLSM

- Each branch receives one dedicated address block.
- Inside each branch, subnets are allocated in descending order of host requirements (largest first).
- Different subnet masks (/24, /26, /27, /28) are applied based on actual host needs, minimizing address waste.
- Each branch retains unallocated space for future VLANs (IP phones, CCTV, biomedical devices, additional departments).